FNO Model Data Specification

1. Input Tensor (X)

Shape: [8, Height, Width]

The input consists of 8 Channels representing the physical geometry and wind conditions.

Idx	Name	Description	Normalization	Range (Approx)
0	SDF	Signed Distance Field (dist to nearest wall)	/ 200.0	0.0 - 1.0
1	Bldg_height	Height of the building at this pixel	/ 50.0	0.0 - 1.0
2	Z_relative	Height slice of the simulation (Z coord)	/ 10.0	0.0 - 1.0
3	U_over_Uref	Background wind ratio (inlet profile)	* 2.0	0.2 - 2.0
4	X_local	X dist from building center	/ 500.0	-1.0 - 1.0
5	Y_local	Y dist from building center	/ 500.0	-1.0 - 1.0
6	dir_sin	Sine of Wind Direction	None	-1.0 - 1.0
7	dir_cos	Cosine of Wind Direction	None	-1.0 - 1.0

2. Target Tensor (Y)

Shape: [1, Height, Width]

The model predicts a single scalar value representing the Wake Deficit.

Target_Delta_U = (Mag_U - U_ref) / U_ref

Interpretation:

- 0.0: No change (Wind speed = U_ref)
- -0.5: 50% drop in speed (Strong Wake)
- -1.0: Stagnation (Speed = 0)

Reconstruction Formula:

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Mag_U = (Prediction * U_ref) + U_ref
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